

Ders Tanıtım Bilgileri						
Ders Kodu	Ders Adı	Yarıyıl	Teorik	Pratik	Kredi	AKTS
BIM304	Makine Öğrenmesi	Güz	3	0	3	5
Temel Bilgiler						
Öğretim Dili:	İngilizce					
Dersin Türü:	Zorunlu					
Dersin Seviyesi:	Lisans					
Dersin Veriliş Şekli:	Yüz yüze					
Dersin Koordinatörü:	Dr. Öğr. Üyesi Alper Öner					
Dersi Veren(ler):	Dr. Öğr. Üyesi Alper Öner, Araş. Gör. Yazım Beril Uluer					
Dersin Amacı:	Makine öğrenmesi dersinin amacı, bilgisayar sistemlerinin verilerden öğrenme yeteneği kazanmasını ve bu öğrenmeleri kullanarak karmaşık problemleri çözebilmesini sağlamaktır. Makine öğrenmesi, yapay zeka alanında önemli bir alt disiplindir ve bilgisayarların veri analizi, desen tanıma, tahmin yapma ve otomatik karar verme gibi görevleri gerçekleştirebilmesini hedefler.					
Dersin İçeriği:	Denetimli Öğrenme, Lineer Cebir-Olasılık, Bayes Anlamlandırma-Karar Teorisi, ML Python Uygulama Kütüphaneleri, Olasılık Tahminleme-Regresyon, Boyut İndirgeme, Kümeleme, Kernel Makineleri, Karar Ağaçları, Sinir Ağları, Bayes Ağları, Saklı Markov Modelleri					
Öğrenme Kazanımları						
Bu dersi başarıyla tamamlayabilen öğrenciler;						
1) Makine öğrenmesi temel kavramlarını ve prensiplerini öğretmek.						
2) Öğrencilerin veri analizi ve özellik mühendisliği becerilerini geliştirmek.						
3) Farklı makine öğrenmesi algoritmalarını anlamalarını ve uygulamalarını sağlamak.						
4) Öğrencilerin model seçimi, model eğitimi ve değerlendirme konularında yetkinlik kazanmalarını sağlamak.						
5) Gerçek dünya problemlerini çözmek için makine öğrenmesi yöntemlerini kullanma becerisi kazandırmak.						
Haftalık Ayrıntılı Ders İçeriği						
Hafta	Konu				Ön Hazırlık	
1)	Makine Öğrenmesine Giriş					
2)	Denetimli Öğrenme					
3)	Bayesian Anlamlandırma - Karar Verme Teorisi					
4)	Çapraz Doğrulama					
5)	Maximum İhtimal Kestirimi - Regresyon					
6)	Boyutsal İndirgeme (PCA, LDA)					
7)	Kümeleme (K-Means ve GMM)					
8)	Ara Sınav					
9)	Kernel Makineleri (SVM)					
10)	Karar Ağaçları (Toplama-Arttırma)					
11)	Yapay Sinir Ağları					
12)	Bayes Ağları					
13)	Saklı Markov Modelleri					
14)	Makine Öğrenmesi Uygulamaları					
Kaynaklar						
Kitaplar:	Introduction to Machine Learning, Ethem Alpaydın, MIT Press, 2016.					
Diğer Kaynaklar:	The Hundred-page Machine. Learning, Andriy Burkov, 2019					
Ölçme ve Değerlendirme Sistemi						

Yarıyıl İçi Çalışmaları	Aktivite Sayısı	Katkı Payı				
Ödev	2	%20				
Proje	2	%40				
Kısa Sınav						
Ara Sınav						
Final	1	%40				
Toplam		%80				
YARIYIL İÇİ ÇALIŞMALARININ BAŞARI NOTU KATKISI		%60				
YARIYIL SONU ÇALIŞMALARININ BAŞARI NOTUNA KATKISI		%40				
Toplam		%100				
AKTS / İş Yüğü Tablosu						
Aktiviteler	Aktivite Sayısı	Süre (Saat)	İş Yüğü	AKTS		
Ders Saati	13	3	39			
Laboratuvar Saati			0			
Sınıf Dışı Çalışma	13	1	13			
Ödev	2	16	32			
Proje	2	15	30			
Kısa Sınav			0			
Ara Sınav			0			
Final	1	20	20			
Toplam İş Yüğü			134	5		
Dersin Öğrenme Kazanımlarının Program Çıktıları ile İlişkisi						
	Etkisi Yok	1 En Düşük	2 Orta	3 En Yüksek		
	Dersin Program Çıktılarına Etkisi					Katkı Düzeyi
1)	Adequate knowledge in mathematics, science, and computer engineering principles, both theoretical and practical, and the ability to apply this knowledge to complex engineering problems.					
2)	Ability to identify, formulate, and solve complex computer engineering problems using appropriate analysis and modeling techniques.					2
3)	Ability to design and develop complex computer systems, devices, or products that meet specific requirements and operate under realistic constraints and conditions, using modern design methods.					2
4)	Ability to develop, select and use modern techniques and tools used for the analysis and solution of complex computer engineering problems, and the ability to use information technologies effectively.					3
5)	Ability to plan and conduct experiments, collect and analyze data, and interpret results in the study of complex computer engineering problems or research topics.					1
6)	Ability to work effectively within and multidisciplinary teams; individual study skills.					2
7)	Ability to communicate effectively orally and in writing; knowledge of at least one foreign language; ability to write effective reports and understand written reports, to prepare design and production reports, to make effective presentations, to give and receive clear and understandable instructions.					1
8)	Awareness of the necessity of lifelong learning; ability to access information, to follow developments in science and technology and to renew continuously.					
9)	To act in accordance with ethical principles, professional and ethical responsibility; information on the standards used in engineering applications.					
10)	Information on business practices such as project management, risk management and change management; awareness of entrepreneurship and innovation; information about sustainable development.					3
11)	Knowledge of the effects of computer engineering practices on health, environment and safety in the universal and social scale and the problems of the era reflected in computer engineering; awareness of the legal consequences of computer engineering solutions.					
Course Introduction Information						

Course Code	Course Title	Semester	Theoretical	Practical	Credit	ECTS								
BIM304	Machine Learning	Fall	3		3	5								
Basic Information														
Language of Instruction:	English													
Type of Course:	Compulsory													
Course Level:	Undergraduate													
Mode of Delivery:	Face to face													
Course Coordinator :	Assist. Prof. Dr. Alper Öner													
Course Lecturer(s):	Assist. Prof. Dr. Alper Öner, Res. Assist. Yazım Beril Uluer													
Course Objectives:	The aim of the machine learning course is to enable computer systems to gain the ability to learn from data and to solve complex problems using these learnings. Machine learning is an important sub-discipline in the field of artificial intelligence and aims to enable computers to perform tasks such as data analysis, pattern recognition, prediction and automatic decision making.													
Course Content:	Supervised Learning, Linear Algebra - Probability, Bayesian Inference - Decision Theory, Python, NumPy, Pandas, Matplotlib, Seaborn, Maximum Likelihood Estimation - Regression, Dimensionality Reduction, Clustering, Kernel Machines, Decision Trees, Neural Networks, Bayes Networks, Hidden Markov Models													
Learning Outcomes														
The students who have succeeded in this course;														
1) To teach the basic concepts and principles of machine learning.														
2) To develop students' data analysis and feature engineering skills.														
3) To enable them to understand and apply different machine learning algorithms.														
4) To enable students to gain competence in model selection, model training and evaluation.														
5) To gain the ability to use machine learning methods to solve real world problems.														
Weekly Detailed Course Contents														
Week	Subject				Related Preparation									
1)	Introduction to Machine Learning													
2)	Supervised Learning													
3)	Bayesian Inference - Decision Theory													
4)	Cross Validation													
5)	Maximum Likelihood Estimation - Regression													
6)	Dimensionality Reduction (PCA, LDA)													
7)	Clustering (K-Means, and GMM)													
8)	Midterm Exam													
9)	Kernel Machines (SVM)													
10)	Decision Trees (Bagging-Boosting)													
11)	Neural Networks													
12)	Bayes Networks													
13)	Hidden Markov Models													
14)	Machine Learning Applications													
Sources														
Textbooks:	Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2016.													
Other Resources:	The Hundred-page Machine. Learning, Andriy Burkov, 2019													
Measurement and Evaluation System														

Semester Requirements	Number of Activities	Weighting													
Homework	2	20%													
Project	2	40%													
Laboratory	-	-													
Quiz															
Midterm Exam															
Final Exam	1	40%													
Total		100%													
PERCENTAGE OF SEMESTER WORK		60%													
PERCENTAGE OF FINAL WORK		40%													
Total		100%													
ECTS / Workload Table															
Activities	Number of Activities	Duration (Hours)	Workload	ECTS											
Course Hours	13	3	39												
Laboratory Hours			0												
Study Hours	13	1	13												
Homework	2	16	32												
Project	2	15	30												
Quiz			0												
Midterm Exam			0												
Final Exam	1	20	20												
Total Workload			134	5											
Correlation between Programme and Learning outcomes															
	No Effect	1 Lowest	2 Average	3 Highest											
	Program Outcomes										Contribution Level				
1)	Adequate knowledge in mathematics, science, and computer engineering principles, both theoretical and practical, and the ability to apply this knowledge to complex engineering problems.														
2)	Ability to identify, formulate, and solve complex computer engineering problems using appropriate analysis and modeling techniques.										2				
3)	Ability to design and develop complex computer systems, devices, or products that meet specific requirements and operate under realistic constraints and conditions, using modern design methods.										2				
4)	Ability to develop, select and use modern techniques and tools used for the analysis and solution of complex computer engineering problems, and the ability to use information technologies effectively.										3				
5)	Ability to plan and conduct experiments, collect and analyze data, and interpret results in the study of complex computer engineering problems or research topics.										1				
6)	Ability to work effectively within and multidisciplinary teams; individual study skills.										2				
7)	Ability to communicate effectively orally and in writing; knowledge of at least one foreign language; ability to write effective reports and understand written reports, to prepare design and production reports, to make effective presentations, to give and receive clear and understandable instructions.										1				
8)	Awareness of the necessity of lifelong learning; ability to access information, to follow developments in science and technology and to renew continuously.														
9)	To act in accordance with ethical principles, professional and ethical responsibility; information on the standards used in engineering applications.														
10)	Information on business practices such as project management, risk management and change management; awareness of entrepreneurship and innovation; information about sustainable development.										3				
11)	Knowledge of the effects of computer engineering practices on health, environment and safety in the universal and social scale and the problems of the era reflected in computer engineering; awareness of the legal consequences of computer engineering solutions.														